vitamin B. Some cofactors are metal ions including iron (Fe), copper (Cu), magnesium (Mg), manganese (Mn), Zinc (Zn), calcium (Ca) and cobalt. If the cofactor is tightly or firmly attached to the apoenzyme it is called a prosthetic group. The prosthetic group may be organic [such as vitamin, sugar, and lipid] or inorganic [such as metal ion] but is not composed of amino acids.

The complete enzyme consisting of the apoenzyme and its cofactor is called the holoenzyme.

4.10.3 Microbial Enzymes

Many microbes synthesize and excrete large quantities of enzymes into the surrounding medium. Using this feature of these tiny organisms many enzymes like Amylase, Cellulase, Catalase, Protease, and Lipase are produced commercially.

Microbial enzymes are extensively used in food processing, preservation, washing powder preparation, leather industry, and paper industry and in scientific research (Table 4.4).

Table 4.4: Industrial application of microbial enzymes

Industries	Enzymes	Microbial Sources	Application
Pharmaceutical industry	• Glucose oxidase • Streptokinase • Protease • coagulase	• *Penicillium notatum* • *Streptococci* • *Clostridium spp* • *Staphylococcus aureus*	• To detect free glucose level in diabetic patients • Anti coagulants • Conversion of fibrinogen to fibrin
Dairy Industry	• Catalase • Lactase	• *Aspergillus niger* • *Lactobacillus spp*	• Remove Hydrogen peroxide in milk (detoxification) • Increase sweetness in milk
Baking Industry	• Amylase • Lipase	• *Bacillus subtilis* • *Candida Lipolytica*	• Increase bread shelf life • Enhances flour quality and dough stability
Polymer Industry	• Lipase • Peroxidase	• *Candida spp* • *Pseudomonas spp*	• Polyester preparation • Formation of cross links
Leather Industry	• Protease Lipase	• *Bacillus spp* • *Aspergillus spp*	• Unbarring of hides, degreasing and softening of leather
Textile Industry	• Cutinase • Collagenase • Laccase	• *Pseudomonas spp* • *Clostridium histolyticum*	• Cotton Scouring • Wool Finishing • Bleach termination • Fabric dyeing
Recombinant DNA technology	• DNase • Ligase	• *Escherichia coli* • *Actinomycetes*	• Nuclease enzyme that break phosphodiester bond of DNA or RNA • Joins the nick in DNA fragments

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